

02225 DRTS: Distributed Real-Time Systems

AVB's Worst-Case Response Time Analysis in TSN

TSN types of traffic

Scheduled Traffic	Audio-Video Bridging (AVB) Traffic	Best Effort (BE) Traffic
+ high determinism	+ real-time capabilities	+ low impact on higher priority traffic
+ low jitter	+ high adaptability	
- low adaptability	- high jitter	- No real-time guarantees

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In summary, TSN defines **three traffic classes**: **Scheduled Traffic (ST)**, **Audio/Video Bridging (AVB)**, and **Best Effort (BE)**. Each of them presents distinct advantages and limitations.

Scheduled Traffic (ST) provides **very high determinism** with **zero jitter**, making it suitable for the most time-critical applications. However, it offers **low adaptability**, since computing an ST schedule is a **computationally expensive task**, as it corresponds to an **NP-hard problem**. In addition, TSN provides **limited support for online TAS reconfiguration**, which further reduces flexibility.

Audio/Video Bridging (AVB) offers **real-time capabilities** by enforcing **bounded delays** and is **highly adaptable**. It can efficiently handle **periodic and aperiodic traffic** thanks to its **online scheduling**, as well as the **dynamic addition or removal of traffic flows**. Nevertheless, AVB **cannot guarantee very low jitter**, which limits its suitability for the most stringent real-time requirements.

Best Effort (BE) traffic has a **low impact on higher-priority traffic classes**, but it **does not provide any real-time guarantees** in terms of

delay or jitter.

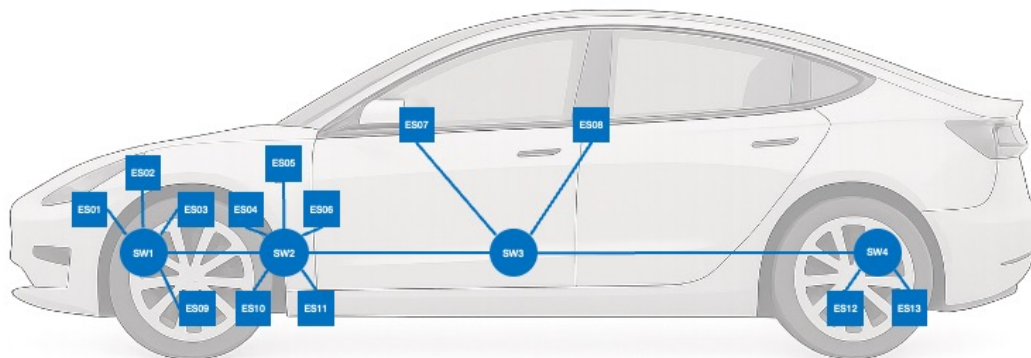
Together, these three traffic classes allow TSN to balance **determinism, flexibility, and efficiency**, depending on the requirements of the application.

Agenda

- System model
 - Network model
 - Traffic model
- Analysis

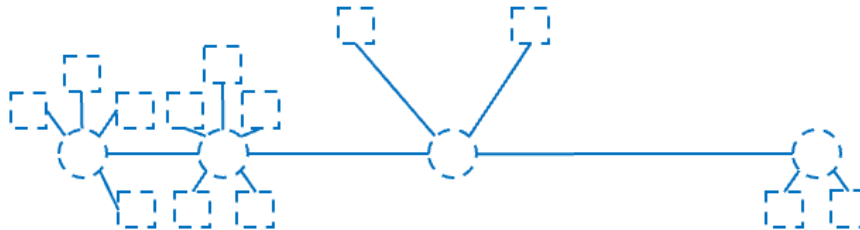
In the following slides, we will introduce the **network and traffic model** that will be used as the basis for our subsequent **analysis of TSN**.

System Model: Network Model



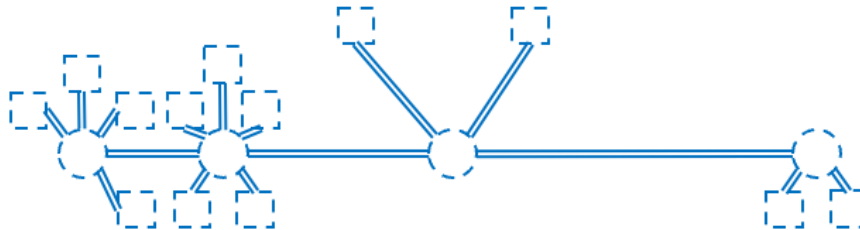
Regarding the **network model**, it is sufficient to focus on the **communication links**, taking into account that links between network nodes are **full-duplex**. Therefore, each full-duplex link is modeled as **two unidirectional links with opposite directions**.

System Model: Network Model



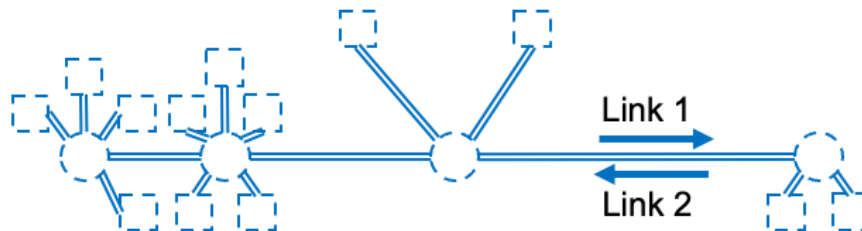
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System Model: Network Model

For all links

$$\forall l_i \in \mathbb{L}$$

Define Bandwidth

$$BW_i$$

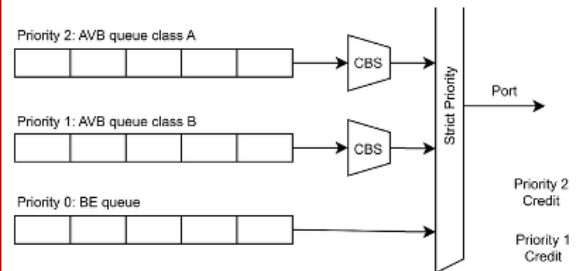
And for every AVB queue in the link

$$\forall P_j^i \in AVB$$

Define sending and idle slope

$$\alpha_j^{i-}$$

$$\alpha_j^{i+}$$



On the other hand, in this presentation—as well as in **Mini-Project 2**—we will analyze a **simplified version of TSN** that **does not include the TAS** and supports **only three priority levels**. The **two highest priorities** are dedicated to **AVB traffic** and therefore implement the **CBS**.

Under this model, for **each link in the network**, we consider the **output port that feeds it**. As a result, each link is characterized by its **bandwidth**. In addition, for each **AVB queue** at the corresponding output port, we define the **rates at which credit is consumed and replenished**, known as **sendSlope** and **idleSlope**, respectively.

System Model: Network Model

$$\alpha_j^{i-} + \alpha_j^{i+} = 1$$

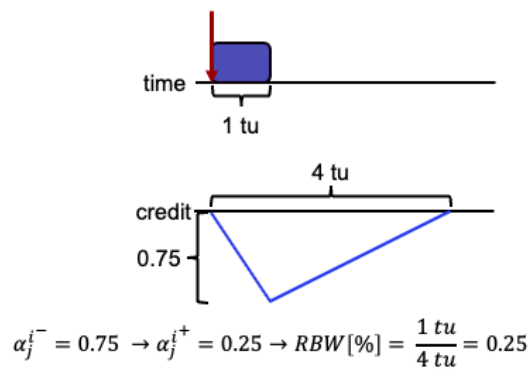
$$\alpha_j^{i+} \times 100 = \text{Percentage of reserved bandwidth(RBW[\%])}$$

Although there are multiple ways to define **sendSlope** and **idleSlope**, one of the most common approaches is to choose them so that **they sum to 1**. With this convention, the value of **idleSlope** directly corresponds to the **fraction of link bandwidth reserved** for that AVB priority on that link.

System Model: Network Model

$$\alpha_j^{i-} + \alpha_j^{i+} = 1$$

$$\alpha_j^{i+} \times 100 = \text{Percentage of reserved bandwidth (RBW[\%])}$$



EXAMPLE

System Model: Network Model

$$\alpha_j^{i-} + \alpha_j^{i+} = 1$$

$$\alpha_j^{i+} \times 100 = \text{Percentage of reserved bandwidth(RBW[\%])}$$

The following analysis is valid for P_j on link l only if:

$$\sum_{\forall P_i | P_i \geq P_j} RBW_i^l \leq 1$$

Therefore, I recommend computing the **idleSlope** for P_j on link l as follows:

$$\alpha_j^{i+} = \frac{U_j^l}{\sum_{\forall P_i \in AVB} U_i^l} \times (1 - U_{BE}^l)$$

However, it is important to note that the **analysis presented next** is only valid when the **sum of the bandwidth reserved by the priority under analysis and all higher priorities is less than or equal to 1**.

For this reason, I recommend distributing the **remaining bandwidth**, after subtracting the utilization of **Best Effort (BE)** traffic, **proportionally to the utilization of the AVB priorities**. This weighted allocation ensures that the assumptions required for the analysis are satisfied while preserving fairness among AVB traffic classes.

System Model: Traffic Model

$$\Gamma = \{m_i (C_i, T_i, D_i, P_i, \mathcal{L}_i) \mid i = 1 \dots N\}$$

- $C_i \rightarrow$ Transmission time ($frame.size/BWD$)
- $T_i \rightarrow$ Period
- $D_i \rightarrow$ Deadline
- $P_i \rightarrow$ Priority
- $\mathcal{L}_i = \{\mathcal{L}_i(0) = l_i, \dots, \mathcal{L}_i(n-1) = l_k\} \rightarrow$ path
 - $n \rightarrow$ number of links in the path
 - $l_i, l_k \in \mathbb{L} \rightarrow$ links in path

On the other hand, traffic is modeled as **flows**, referred to as **streams**. A stream represents a sequence of frames that share similar characteristics, such as **frame size**, **periodicity**, and other traffic parameters.

Streams can be seen as the **communication equivalent of tasks** in real-time systems, where **each frame corresponds to a job or task instance** of the stream.

Streams are characterized by several parameters. First, their **transmission time**, which is computed by dividing the **frame size** by the **link bandwidth**. In addition, each stream is defined by its **period**, **deadline**, and **priority**.

Finally, streams are associated with a **path**, which represents the **set of links in the network model** that the stream must traverse in order to travel from the **source end-system to the destination end-system**.

Agenda

- System model
 - Network model
 - Traffic model
- Analysis:
 - **Worst-Case Response Time Analysis (WCRTA)**
based on **eligible interval**

The analysis presented next is a **Worst-Case Response Time (WCRT) analysis** based on the concept of **eligible intervals**. More detailed explanations, as well as formal proofs of this analysis, can be found in the paper referenced in **Mini-Project 2**.

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} WCRT_i^l$$

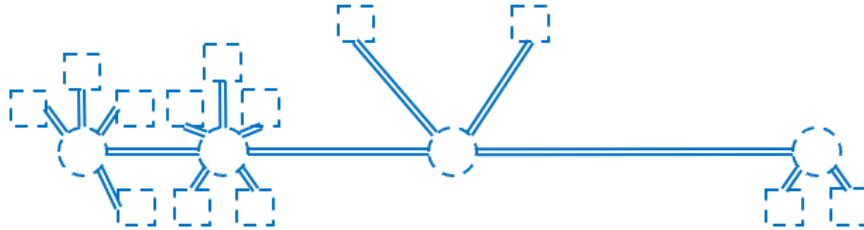
First, this analysis is **compositional**, meaning that it computes the **Worst-Case Response Time (WCRT)** from the **source to the destination** by **summing the WCRT contributions of each link** along the stream's path.

Other parameters, such as **switch delay**—that is, the time elapsed between a frame being received at an input port and being enqueued at the corresponding output port—or **propagation delay**, namely the time required for a frame to travel across a link, are assumed to be **negligible** when compared to the **transmission time**. The transmission time is defined as the interval between the moment a frame enters the transmission queue and the moment it is fully transmitted onto the link.

When necessary, these additional delays can be incorporated into the analysis as **constant offsets**, depending on the specific characteristics of the switch or the link.

Analysis

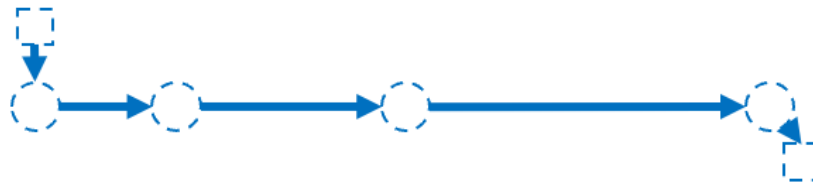
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$$WCRT_i^l = SPI_i^l + HPI_i^l + LPI_i^l + C_i$$

On the other hand, the **Worst-Case Response Time (WCRT)**—or, in this context, the **transmission time**—is computed by taking into account **three types of interference**, in addition to the transmission time of the frame under analysis itself.

Same-Priority Interference (SPI): interference caused by frames with the **same priority** as the frame under analysis.

Higher-Priority Interference (HPI): interference caused by frames with **higher priority** than the frame under analysis.

Lower-Priority Interference (LPI): **blocking** caused by frames with **lower priority**, which may delay the transmission of the frame under analysis.

Analysis

$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$

Priority 2: AVB queue class A



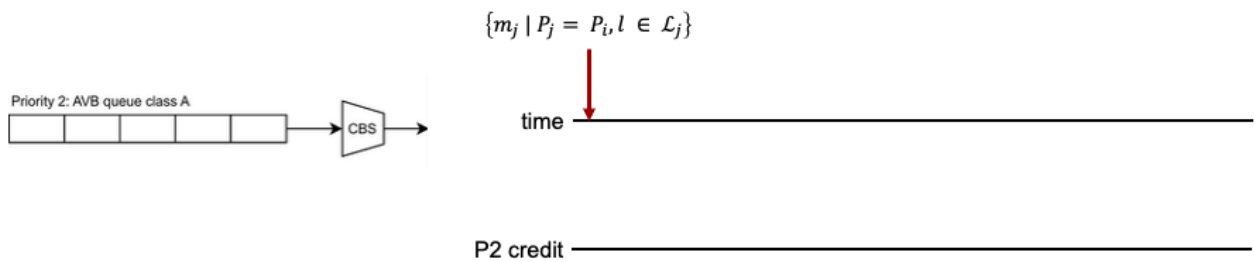
time _____

P2 credit _____

Let us begin by analyzing the **Same-Priority Interference (SPI)** for a **single AVB queue**.

Analysis

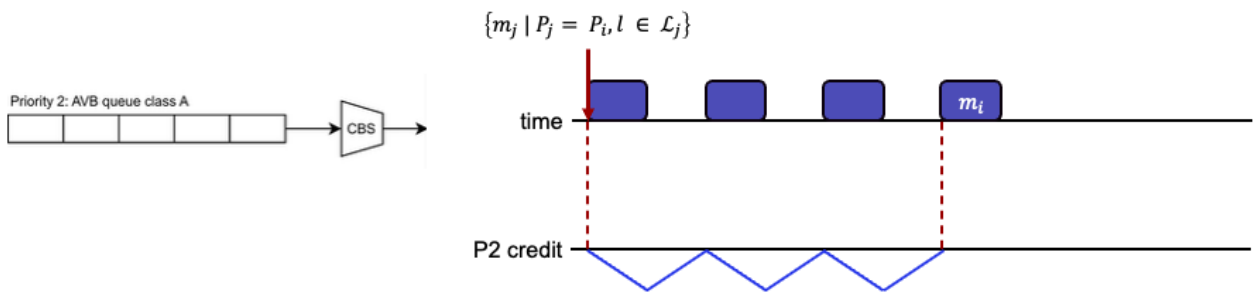
$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$



When a frame arrives at the **transmission queue**, if the **deadline of the streams is less than or equal to their period**, then at most **one frame per stream** with the **same priority** as the frame under analysis can be present in the queue. Otherwise, one of the frames already in the queue would have **missed its deadline**.

Analysis

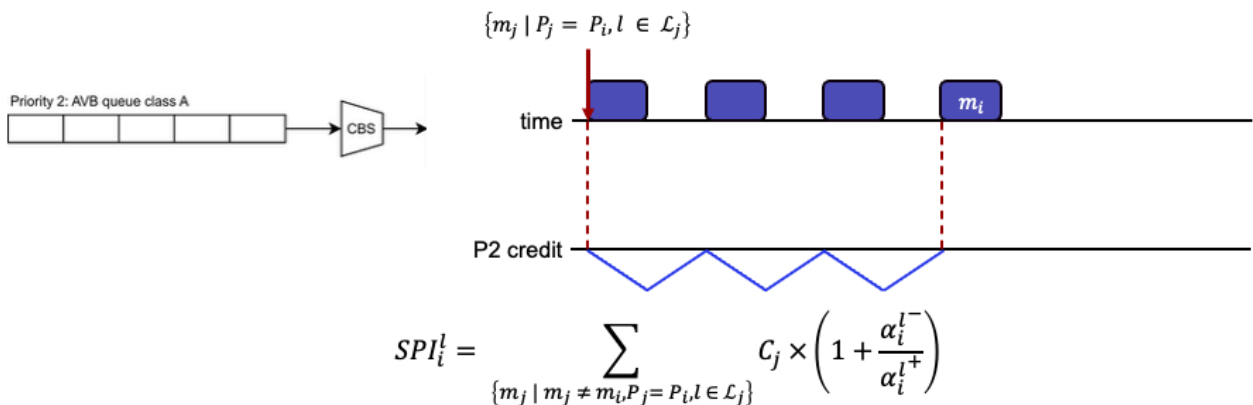
$$WCRT_i^l = \boxed{SPI_i^l} + HPI_i^l + LPI_i^l + C_i$$



Therefore, for the frame under analysis to be transmitted, **all frames ahead of it in the queue** must be transmitted first. However, it is not sufficient to simply sum the **transmission times of the same-priority frames**. It is also necessary to account for the **time required to recover the credit** consumed by each of those transmissions.

Analysis

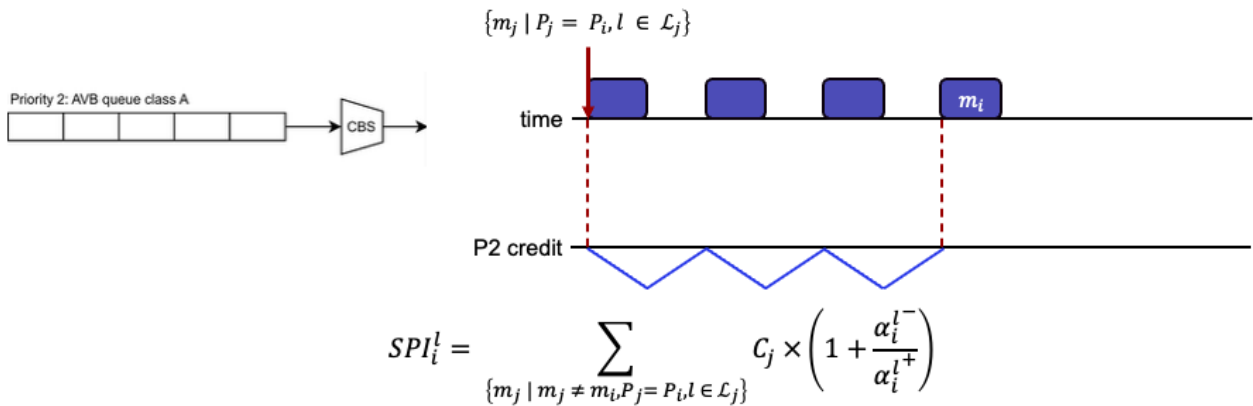
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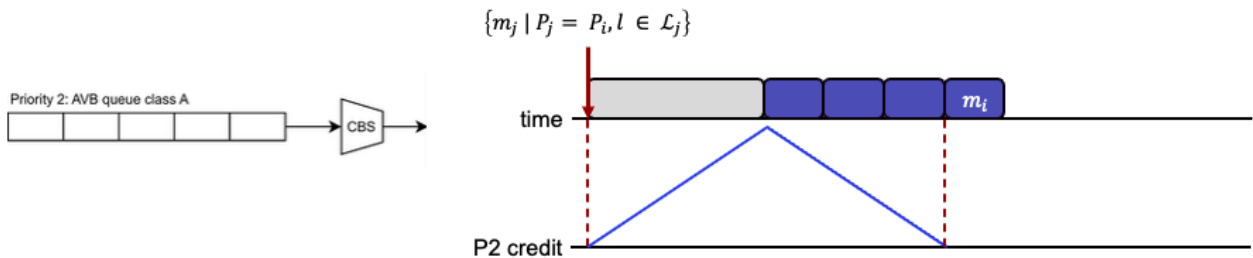
Next, we will compute the **WCRT components corresponding to higher- and lower-priority interference**. Before doing so, it is important to note the following observation.

If the **credit level is the same at the beginning and at the end of the analysis interval**, then it is irrelevant whether interference from other sources has occurred or not. This is because any such interference is effectively reflected as **credit recovery time**, which has already been accounted for in the **Same-Priority Interference (SPI)** term.

Consequently, this interference **must not be added again**, as doing so would result in double counting.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$



$$SPI_i^l = \sum_{\{m_j \mid m_j \neq m_i, P_j = P_i, l \in \mathcal{L}_j\}} C_j \times \left(1 + \frac{\alpha_i^{l-}}{\alpha_i^{l+}}\right)$$

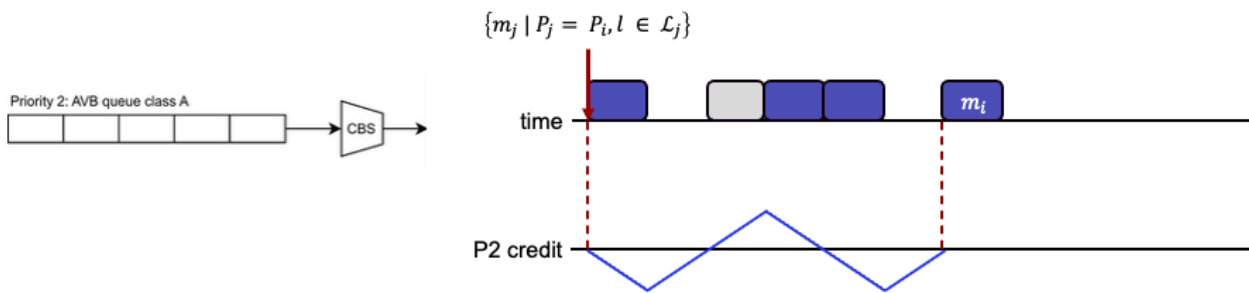
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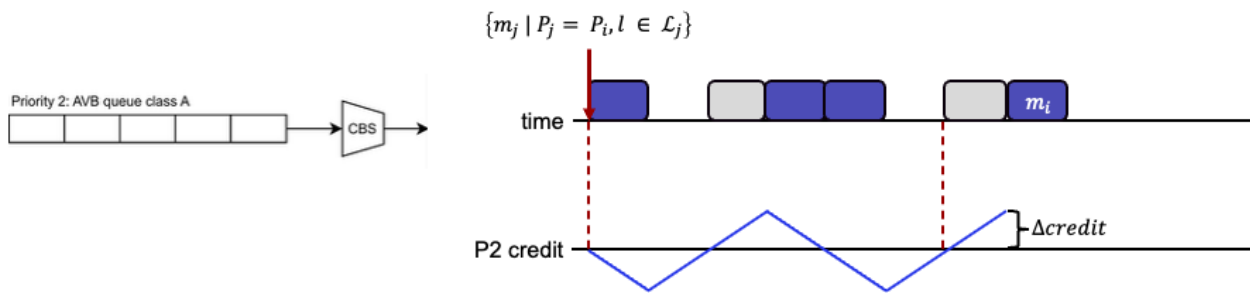
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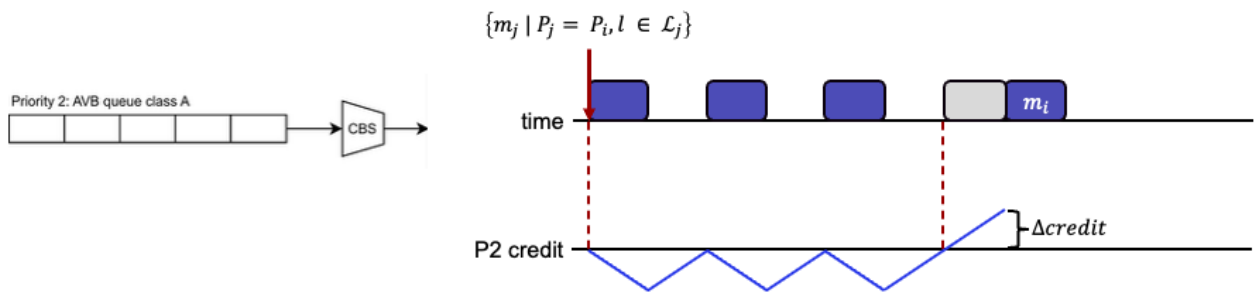


In this sense, any **additional delay beyond the Same-Priority Interference (SPI)** will be reflected as a **positive difference in credit** between the beginning and the end of the analysis interval.

Therefore, if we can determine the **maximum credit that can be accumulated for each priority**, we can compute the **Higher- and Lower-Priority Interference** by calculating the **time required to reach that credit level**.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

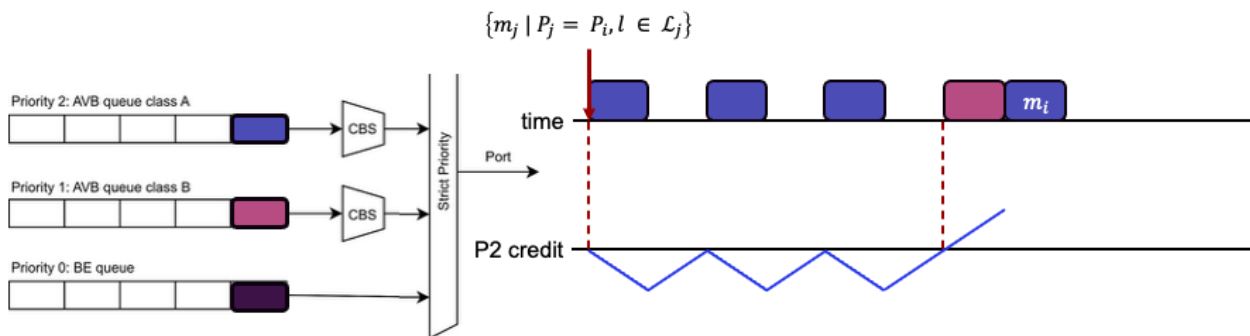


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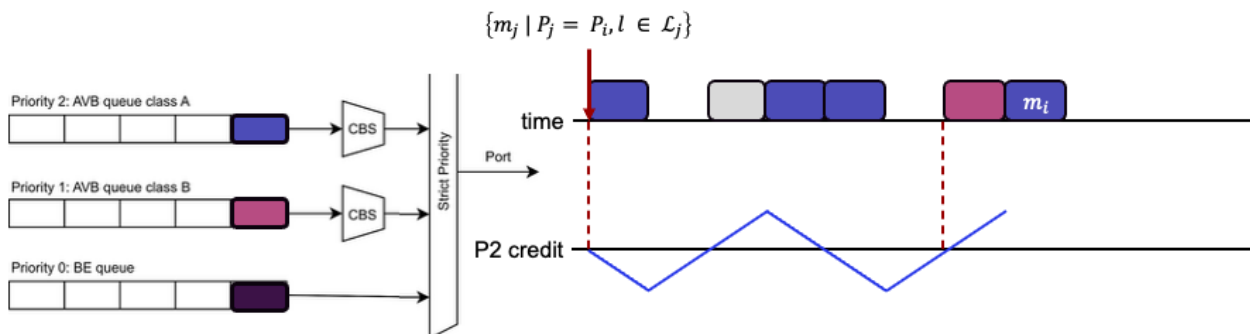


Let us start by analyzing the **highest-priority AVB queue or class A**. On the one hand, this queue can only be **blocked by lower-priority traffic**, since there is **no higher-priority traffic** that could interfere with it. Moreover, blocking can occur **only when its credit is negative**, because otherwise this queue would win access to the link due to its highest priority.

As a result, the **maximum credit** is reached when the queue has **negative credit**, and **just before the credit recovers to zero**, a **lower-priority frame with the largest transmission time** starts transmitting.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$



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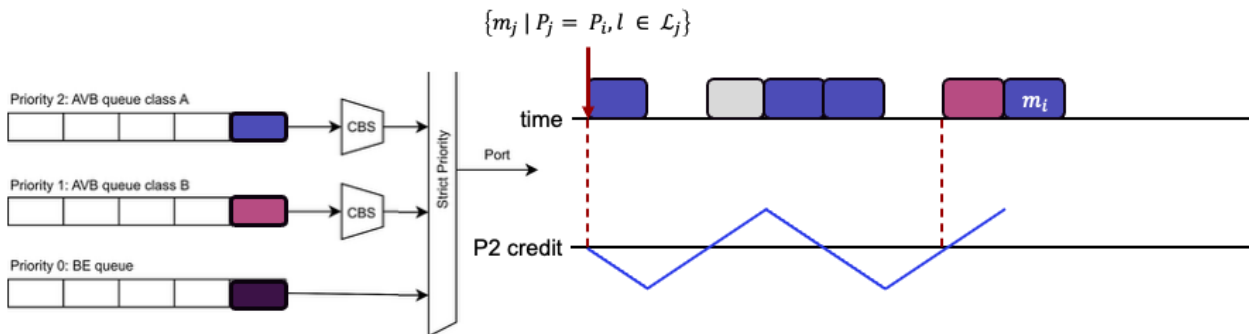
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As a result, the **maximum credit** is reached when the queue has **negative credit**, and **just before the credit recovers to zero**, a **lower-priority frame with the largest transmission time** starts transmitting.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

0 $\max_{m_j | P_j < P_i, l \in \mathcal{L}_j} C_j$

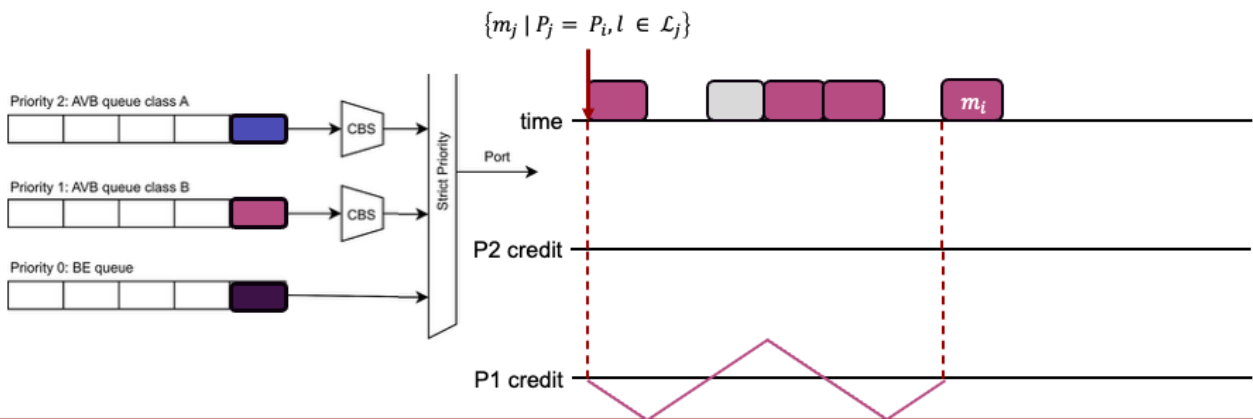


Under these conditions:

- **Higher-Priority Interference (HPI) = 0**, since no higher-priority traffic exists.
- **Lower-Priority Interference (LPI)** is equal to the **maximum transmission time among all lower-priority streams**.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

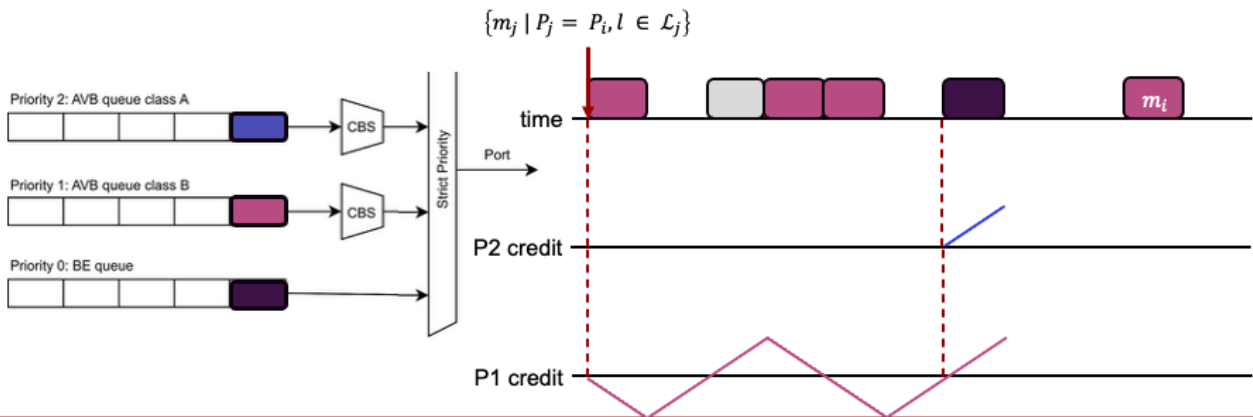


For AVB Class B traffic, the **Lower-Priority Interference (LPI)** is the same as for AVB Class A, under the same conditions described previously.

Analysis

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

$\max_{m_j | P_j < P_i, l \in \mathcal{L}_j} C_j$



However, in this case, an additional source of interference must be considered. During the blocking caused by lower-priority traffic, the **credit of the higher-priority AVB Class A queue also increases**. As a consequence, in addition to LPI, we must now account for **Higher-Priority Interference (HPI)**.

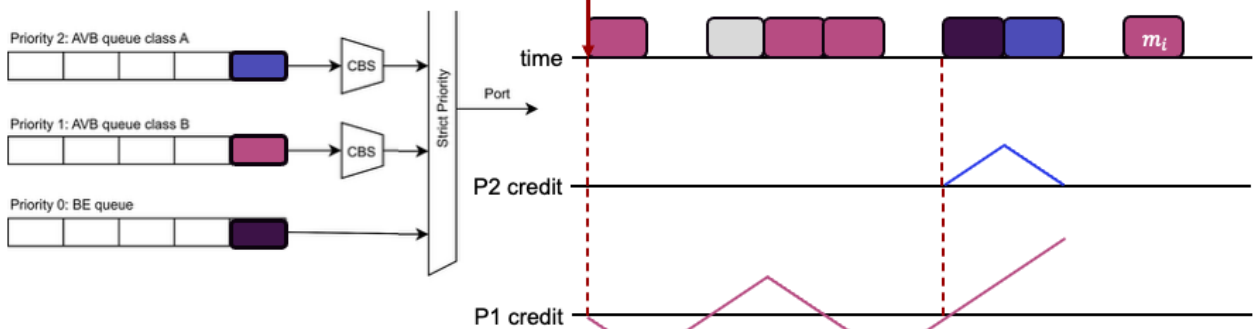
Analysis

$$LPI_i^l \times \frac{\alpha_H^{l+}}{\alpha_H^{l-}}$$

$$m_j \mid P_j < P_i, l \in \mathcal{L}_j \quad \max C_j$$

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

$$\{m_j \mid P_j = P_i, l \in \mathcal{L}_j\}$$



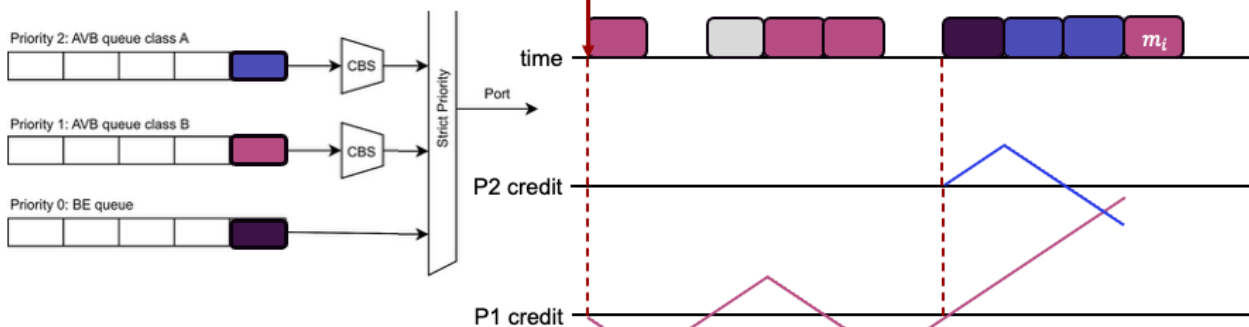
In the **worst-case scenario**, the accumulated **Class A credit**—which built up during the blocking interval—is **fully consumed first**.

Analysis

$$LPI_i^l \times \frac{\alpha_H^l +}{\alpha_H^l} + \max_{m_j | P_j > P_i, l \in \mathcal{L}_j} C_j$$

$$WCRT_i^l = SPI_i^l + \boxed{HPI_i^l + LPI_i^l} + C_i$$

$$\{m_j | P_j = P_i, l \in \mathcal{L}_j\}$$



Then, **just when the Class A credit reaches zero**, the transmission of the **largest frame among the higher-priority (Class A) streams** begins.

Therefore, for AVB Class B traffic:

- **LPI** is identical to that of AVB Class A.
- **HPI** consists of the time required to consume the accumulated **Class A credit**, plus the **maximum transmission time** of the higher-priority streams.

Analysis

$$WCRT_i = \sum_{\forall l \in \mathcal{L}_i} WCRT_i^l$$

$$WCRT_i^l = SPI_i^l + HPI_i^l + LPI_i^l + C_i$$

$$SPI_i^l = \sum_{\{m_j \mid m_j \neq m_i, P_j = P_{i,l} \in \mathcal{L}_j\}} C_j \times \left(1 + \frac{\alpha_i^{l-}}{\alpha_i^{l+}}\right)$$

$$LPI_i^l = \max_{m_j \mid P_j < P_{i,l} \in \mathcal{L}_j} C_j$$

AVB Class A

$$HPI_i^l = 0$$

AVB Class B

$$HPI_i^l = LPI_i^l \times \frac{\alpha_H^{l+}}{\alpha_H^{l-}} + \max_{m_j \mid P_j > P_{i,l} \in \mathcal{L}_j} C_j$$

Finally, this slide **summarizes the analytical formulas** used in the **Worst-Case Response Time analysis for AVB traffic**.